MIL-I-22126B(AS)
17 August 1972
Superseding
MIL-I-22126A(WP)
15 December 1964

MILITARY SPECIFICATION

INDICATOR, TORQUEMETER, ENGINE, TURBOPROP

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers design and all performance requirements for procurement of turboprop engine torquemeter indicators for use on P-3 aircraft.

1.2 Classification - The indicators shall be of the following types:

Indicators with power-off flag.

MS90323-1	2-inch, lighted
MS90323-2	3-inch, lighted
MS90323-3	2-inch, non-lighted
MS90323-4	3-inch, non-lighted

Indicators without power-off flag.

MS90323-5	3-inch, lighted
MS90323-6	3-inch, non-lighted

2. APPLICABLE DOCUMENTS

2.1 General - The following documents, of the issue in effect on the date of the invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

BB-N-411	Nitrogen
QQ-C-320	Chromium Plating (Electrodeposited)
QQ-N-290	Nickel Plating (Electrodeposited)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

Federal (Continued)

QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-Z-325	Zinc Plating (Electrodeposited)

Military

MIL-P-116	Preservation, Methods of
MIL-M-3171	Magnesium Alloy, Processes for Corrosion Protection of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-S-6872	Soldering Process, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-C-14806	Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges
MIL-P-23408	Plating: Tin-Cadmium (Electrodeposited)
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

Military (Continued)

MIL-C-26482	Connectors, Electric, Circular, Miniature Quick Disconnect
MIL-B-27497	Bearing, Jewel, Sapphire or Ruby, Synthetic
MIL-G-81704	Glass, Aircraft Instrument, Lighting Wedge and Cover

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-756	Reliability Prediction

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

STANDARDS

Military(Continued)

MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures For Packaging and Packing of
MIL-STD-889	Dissimilar Metals
MS24367	Lamp, Incandescent - Miniature Integral Lighting
MS24515	Lamp, Subminiature
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33585	Pointers, Dial, Standard Design of Aircraft Instrument
MS90323	Indicator, Torquemeter, Engine, Turboprop

PUBLICATIONS

NAVAL AIR SYSTEMS COMMAND

AR-5	Microelectronic Devices Used in Avionics Equipment Procedures for Selection and Approval of
AR-10	Maintainability of Avionics Equipment and Systems, General Requirements for

2.2 Availability of documents - When requesting any of the applicable documents, refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Qualification - The indicators furnished under this specification shall be a product which has been tested and meets the Quality Assurance Provisions specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Selection of government documents - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial parts - Commercial parts having suitable properties shall be used, where on the date of invitations for bids, there are no suitable standard parts. In any case, commercial parts, such as screws, bolts, nuts, cotter pins, having suitable properties, may be used provided:

(a) They can be replaced by the standard parts (MS or AN) without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and if practicable, on the contractor's drawings.

3.2.2 Standard parts - With the exception of 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials - Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight and suitable for the purpose intended.

3.3.1 Critical materials - Noncritical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 Non-magnetic materials - Non-magnetic materials shall be used for all parts of the indicator except where magnetic materials are essential or required.

3.3.3 Metals - Metals shall be of the corrosion resistant type, or shall be suitably protected as specified herein to resist corrosion due to fuels, salt spray or atmospheric conditions to which the instrument may be subjected when in storage, during normal service life, and when subjected to the tests specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

3.3.3.1 Dissimilar metals - Dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact with each other, unless suitably protected against electrolytic corrosion by means of protective coatings. Protective coating will not be necessary if the metals are in a sealed atmosphere as described in 3.4.5.2.

3.3.3.2 Magnesium alloy parts - Magnesium alloy parts shall be treated in accordance with MIL-M-3171. When abrasion resistance is a factor, an anodic treatment approved by the procuring activity shall be used.

3.3.3.3 Aluminum alloy parts - Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to Specification MIL-A-8625. Small holes, pipe threads and case inserts need not be anodized. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.3.3.4 Iron and steel parts - Iron and steel parts which are in hermetically sealed cases shall be cadmium plated in accordance with QQ-P-416. Parts which are not in sealed cases shall be chromium, nickel, or zinc plated in accordance with QQ-C-320, QQ-N-290, or QQ-Z-325 respectively. Parts in a confined space in the presence of organic materials shall be tin-cadmium plated in accordance with MIL-P-23408. The class and type of plating shall meet the requirements of 3.3.4.

3.3.4 Protective treatment - When materials are used in the construction of the indicator that are subject to atmospheric or environmental conditions likely to cause corrosion in normal service life, they shall be protected against corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. Finishes and protective coatings which will crack, chip or scale during normal service life or are affected by extremes of atmospheric or environmental conditions, shall not be used.

3.3.5 Fungus-proof materials - Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used, they shall be treated with a fungicidal agent acceptable to the procuring activity. If used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

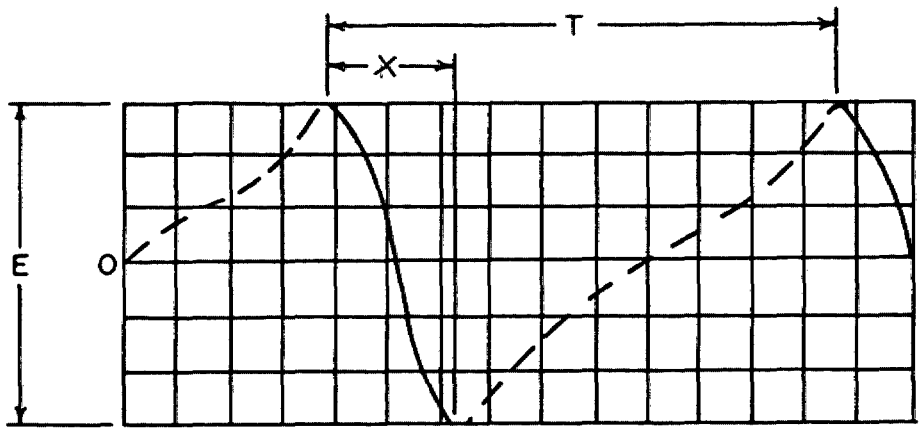
3.3.6 Fumes and vapors - Materials used in the construction of the instrument shall not produce corrosive, deleterious, or toxic fumes or vapors under all conditions specified herein.

3.4 Design and construction -

3.4.1 Signal from the transmitter - The indicator shall accept from the torquemeter transmitter a 2 phase signal whose peak-to-peak voltage is constant at 13820 RPM, throughout the range of phase displacement, the voltage difference between the two phases does not exceed 10 percent of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

larger value, and whose phase difference varies linearly with torque or with indicated horsepower at 13820 RPM*. The input signal waveform (both phases) will be a modified sinusoidal wave with characteristics as shown in Figure 1. The d.c. resistance of either signal source, i.e., either torquemeter pickup coil, is 2.5±0.5 ohms. At 9.2 KHz its impedance is approximately 4.1 ohms. (See 6.6 for cautionary information.)



	RANGE		
FREQUENCY, KHz	2.6	4.6	10.0
T, TOTAL PERIOD, μ SEC	400	218	100
X, NEGATIVE SLOPE PORTION OF TOTAL PERIOD, MICROSECONDS	___	100-150	28-38
E, PEAK TO PEAK AMPLITUDE, VOLTS	0.40,MIN. 0.60,MAX.	0.75,MIN. 6.0,MAX.	0.75,MIN. 6.0,MAX. 1.5,NOM.

Figure 1. Input Signal Waveform

The relationship that exists in the transmitter between electrical degrees of phase displacement and torque is fixed for any particular transmitter but may vary from transmitter to transmitter in the manner described below:

(a) The torque is zero for some phase difference in the interval 45 ± 8.4 degrees.

(b) The slope of the phase difference versus torque line is 17.96 ± 0.27 horsepower per degree of phase difference.

*Hereafter when horsepower is mentioned in this specification, horsepower at 13820 RPM is implied. The electrical output signal from the torquemeter transmitter has a frequency of approximately 9.213 KHz at 13820 RPM engine torque shaft speed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

An alternative way of stating the above is that the signal coming from the transmitter will have a phase difference which varies with horsepower as in the equation below. Furthermore, the variation in this relationship from transmitter to transmitter is as shown in the constants of this equation.

$$\text{Phase Difference} = \frac{\text{Horsepower}}{17.96 \pm 0.27} + 45 \pm 8.4$$

3.4.2 Indicator adjustments - The indicator shall be provided with a zero adjustment, identified as "CAL A", a slope adjustment, identified as "CAL B", and a two pole-triple throw switch.

The adjustments and switch shall be permanently attached to the back of the indicator, shall be operable using only a screwdriver, and shall be constructed and mounted on the indicator so as to preclude inadvertent movement of the adjustments or switch position during handling and normal use.

The two adjustments and switch shall be sufficient for fitting the indicator to the characteristics of a particular transmitter. It must be possible to match the indicator to the torque transmitter of the engine, using only the information in 6.5 and the aforementioned switch and adjustments.

The adjustments shall be provided with stops at either end of the range of each adjustment and shall allow setting within the range of adjustment (see 4.6.3) to within $\pm$ 1 HP.

When the switch is in the center position both the reference and torque signals shall be supplied to the indicator mechanism.

When the switch is in the position identified "A", the reference input signal shall be interrupted. The interruption shall be equivalent to interrupting the signal input to pin D of the indicator.

When the switch is in the position marked "B", the torque signal shall be interrupted. The interruption shall be equivalent to interrupting the signal input to pin B of the indicator.

It shall also be possible to calibrate the indicator utilizing the adjustments in connection with the CAL "A" - CAL "B" switches shown in the interconnecting wiring harness of Figure 2.

3.4.3 Maintenance - The design shall be reasonably simple to facilitate as much as possible disassembly, repair or overhaul, service maintenance, and reassembly using those tools and items of maintenance equipment which are normally available as commercial standards.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

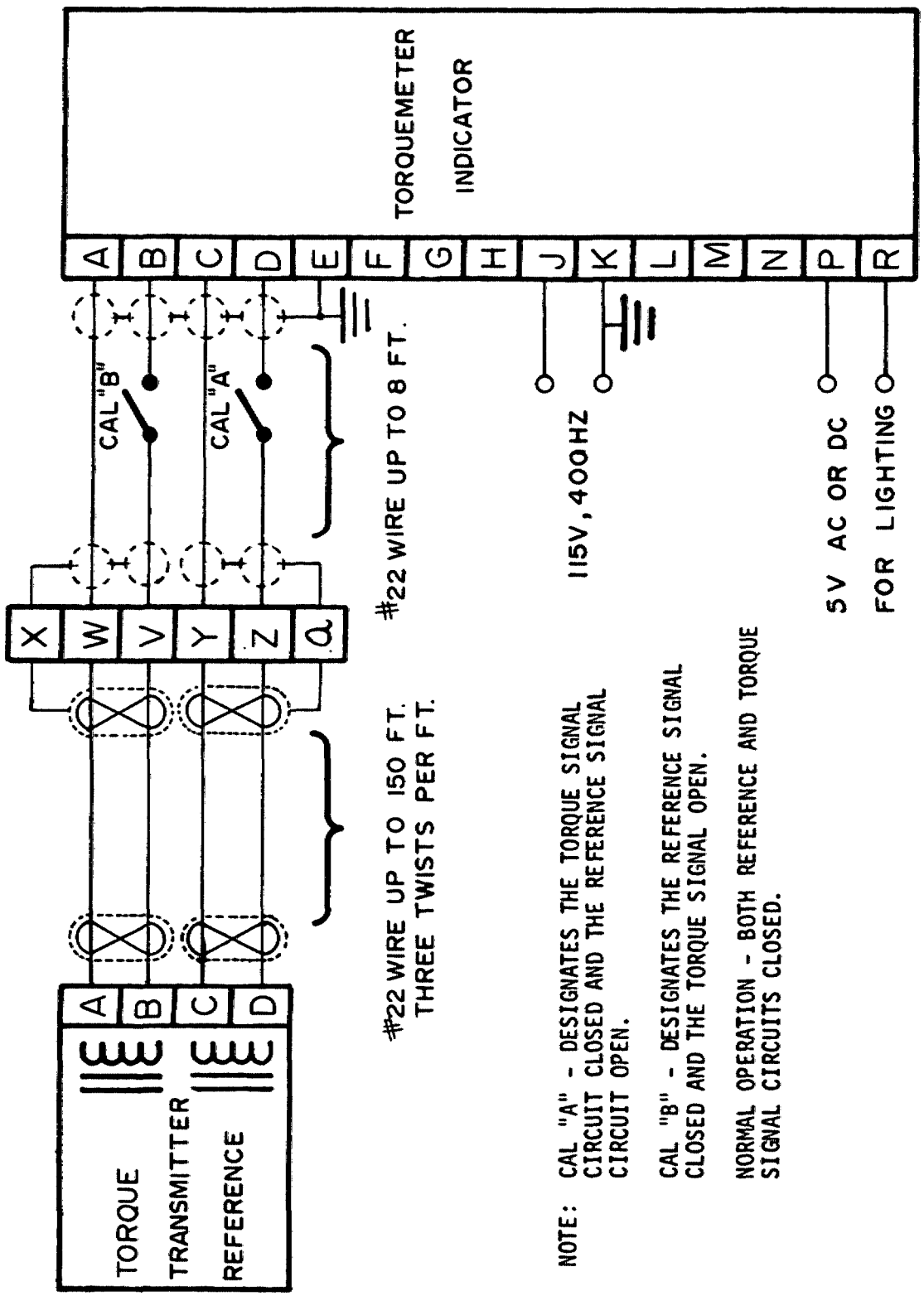


FIGURE 2. WIRING

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

3.4.4 Power-off flag -

Where specified in 1.2, the indicator shall be provided with a power-off flag. The flag, marked with the legend "OFF", shall be visible through the cover glass when the system is de-energized. When the system is energized, a blank sign of the same color as the dial background shall appear in place of the legend.

3.4.5 Case - The case shall conform to MS90323. The case and mounting flange shall be made of non-ferrous, low density metal, uniform in texture, having a smooth surface and shall be hermetically sealed. The case shall be finished with a lusterless black material, Color No. 37038 of FED-STD-595. The finishing material shall be of a durable type to withstand usage encountered in service and the tests specified herein.

3.4.5.1 Hermetic sealing - The case shall provide a hermetically sealed enclosure for all of the mechanism. The case shall be so constructed that it may be opened, the mechanism removed and replaced and the case re-sealed at least three times without case or mechanism damage. This shall be possible without the use of any special tool, jig or fixture, unless such device is specifically approved by the procuring activity. The sealing of the case shall not be dependent upon any material which will be adversely affected by any atmosphere to which the instrument may be subjected in normal use in military aircraft.

3.4.5.2 Filling medium - The filling medium shall be a mixture of 90 percent nitrogen and ten percent helium. The nitrogen used shall be in accordance with BB-N-411, Type I, Class I, Grade C. The filling medium shall contain not more than 0.006 milligram of water vapor per liter (dew point -65°C) at the filling pressure. The absolute pressure of the filling medium in the case shall be 1 $\pm$ 0.1 atmosphere.

3.4.6 Weight - The indicator shall be so designed that when completely assembled the weight shall not exceed 2.0, 2.35 pounds for the 2-inch unlighted and lighted indicators respectively or 3.0 and 3.25 pounds for the respective 3-inch indicators.

3.4.7 Screw threads - Screw threads 0.060 inch in diameter or larger shall be in accordance with MIL-S-7742.

3.5 Electrical requirements -

3.5.1 Primary input power requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

(1) AC Power (Single Phase) 115V, Category "B", 15 VA

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

3.5.2 Undervoltage protection - The indicator shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage is within the specified limits.

3.5.3 Electronic parts - Electronic parts and the application thereof shall be in accordance with MIL-E-5400. Parts that do not appear on approved lists in accordance with MIL-E-5400 shall not be used unless approved by the procuring activity.

3.5.4 Electronic devices -

3.5.4.1 Transistors and diodes shall be provided and applied as outlined in MIL-E-5400. Electron tubes shall not be used.

3.5.4.2 The electronic portions of the equipment shall be modularized in accordance with AR-10.

3.5.4.3 When used, Microelectronic Modular Assemblies shall meet the requirements of AR-5.

3.5.5 Integral lighting - The models specified for integral lighting shall be provided with 5V integral lighting in accordance with MIL-L-25467.

3.5.6 Lamps - The lamp shall be located within the indicator case. Lamps conforming to MS24367 or MS24515 may be used in lieu of, or in addition to, the lamps specified in MIL-L-25467.

3.5.7 Electric connectors - The electric connector used shall conform to MIL-C-26482. The type connector used shall be as shown in MS90323. The connector code number markings as specified in MIL-E-5400 shall be legibly and permanently applied to the case or connector of the indicator.

3.5.8 Wiring - The indicator shall be wired so that it meets all specification requirements when externally connected as shown in Figure 2. Internal wiring shall be color coded, neat and accomplished in such a manner that individual wires may be easily traced.

3.6 Dial and pointer - The dial shall be securely mounted in such a manner that it will not loosen or slip when the indicator is vibrated. If screws are used they shall be so located that they will not interfere with the proper location of any dial marking.

3.6.1 Visibility of dial - The pointer, numerals, at least 1/3 of the shortest graduations and all other specified markings on the dial shall be visible from any point within the frustum of a cone whose side makes an angle of 30° with a perpendicular to the dial and whose small diameter is the aperture of the case.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

3.6.2 Dial marking - The dial shall be marked as shown in Figure 3. Unless otherwise specified, all visible portions of the dial shall be lusterless black Color No. 37038 of FED-STD-595 and all markings on the dial shall be lusterless white Color No. 37875 of FED-STD-595. All markings shall be durable to withstand usage encountered in service. The form of the numerals and letters shall be in accordance with MS33558. Numerals shall distinctly indicate the graduation to which each applies. If practicable, each numeral shall be placed so that the center of area of numeral(s) is on the radial line joining the appropriate graduation and the center of the dial. Any confusion resulting in doubt as to the graduation to which the numeral applies shall be cause for rejection. When several numerals are used in one group, the space between the numerals shall be not less than 1/64th inch.

3.6.3 Pointer - The pointers of the indicator shall be as light weight as practicable, sufficiently rigid to prevent oscillation under vibration and shall be firmly attached to the mechanism but also shall be readily adjustable. The shaded portion of the pointers shall be finished lusterless white, Color No. 37875 of FED-STD-595. The unshaded portion shall be finished lusterless black, Color No. 37038 of FED-STD-595. Pointers shall conform to MS33585-9. The pointers of the lighted indicators shall be suitable for integral lighting.

3.7 Cover glass - The cover glass shall be in accordance with MIL-G-81704 and shall be coated in accordance with MIL-C-14806.

3.7.1 Cover glass mounting - The distance from the inner surface of the cover glass to the surface of the dial on which the marking is applied shall not exceed 0.250 inch.

3.8 Performance - The equipment shall perform satisfactorily when subjected to the tests specified in Section 4 of this specification.

3.8.1 Reliability - Indicator reliability shall be analyzed, calculated, and reported in accordance with MIL-STD-756.

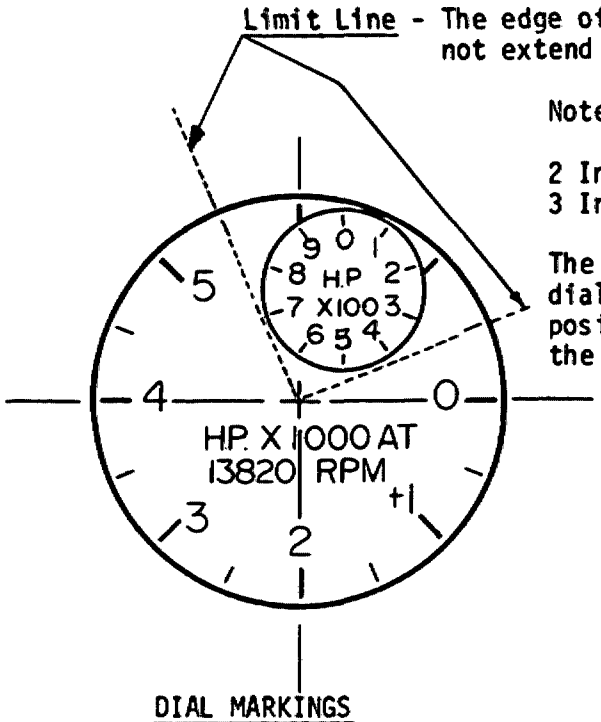
3.8.2 Reliability in mean-time-between-failures - The indicator shall have at least 1500 hours of mean (operating) time between failures, (MTBF), when tested and accepted as outlined under the requirements of 4.4.3.

3.9 Identification of product -

3.9.1 Nameplate - A nameplate shall be securely attached to the exterior of the case and shall be marked in accordance with the requirements of MIL-STD-130, except that the FSN marking shall be omitted. The following statement shall appear on the space reserved for special characteristics:

CAUTION - This indicator must be calibrated to the torquemeter transmitter with which it is used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.



DIAL MARKINGS

	2 inch size		3 inch size
Thickness of all numerals and letters		0.025	
		MAIN DIAL	
Numerals, height	0.156		0.25
Major graduations, length	0.187		0.187
Major graduations, width	0.030		0.032
Minor graduations, length	0.125		0.125
Minor graduations, width	0.025		0.025
Angle between graduations	22-1/2°		22-1/2°
"HP X 1000 at 13820 RPM", height	0.125		0.156
OFF (on power-off flag, not shown)	0.140		0.200
		SUBDIAL	
Numerals, height	0.0625		0.0938
Major graduations, length	0.0938		0.125
Major graduations, width	0.020		0.025
Minor graduations, length	-----		0.0938
Minor graduations, width	-----		0.020
Angle between graduations	36°		18°
"HP." height	-----		0.0938
"X100" height	-----		0.0625
"HP X 1000", height	0.0625		-----

On 2-Inch subdial, odd numerals are optional. Unless otherwise noted all dimensions in inches. Tolerances: 3 place decimals ± 0.005, 4 place decimals ±0.0005. Angles±1/2 degrees.

A power-off flag is not required in either MS90323-5 or MS90323-6 indicator.

Figure 3. Dial

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

3.9.2 Manufacturer's part number - The manufacturer's part number on the nameplate shall be identical with the manufacturer's engineering production drawing number including applicable dash numbers if the drawing is tabulated and covers more than one part.

3.10 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part number shall be governed by the drawing number requirements of MIL-STD-100.

3.11 Installation -

3.11.1 Installation instructions - The contractor shall furnish with each instrument one copy of installation instructions, including calibration procedure, with illustrations and diagrams. Prior to printing, two copies shall be furnished to the procuring activity for approval. Whenever possible the instructions shall be arranged to require only one sheet of paper either 8-1/2 by 11 or 11 by 17 inches. The instrument calibration instructions shall be compatible with the applicable aircraft wiring.

3.11.2 Envelope - An envelope furnished by the contractor shall be packaged with each instrument and shall be marked as follows:

IMPORTANT

THIS ENVELOPE CONTAINS INSTALLATION INSTRUCTIONS

3.12 Soldering - Soldering shall be performed in accordance with MIL-S-6872 and Requirement 5 of MIL-STD-454.

3.13 Workmanship - The indicator including all parts and accessories shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, riveting, machine-screw assemblage, welding and brazing, and freedom of parts from burrs and sharp edges.

3.14 Interference control - The generation of radio interference by the torquemeter system and the vulnerability of the system to radio interference shall meet the requirements of MIL-STD-461.

3.15 Jewel bearings - When used, synthetic sapphire or ruby jewel bearings shall be in accordance with MIL-B-27497.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of Tests - Inspection of the indicators shall be classified as follows:

(a) Qualification Tests: Qualification inspection consists of examinations and tests performed on sample indicators submitted for approval as a qualified product.

(b) Quality conformance inspection: Quality conformance inspection consists of examinations and tests performed on indicators manufactured and submitted for acceptance under contract.

4.3 Qualification inspection - The qualification inspection of the indicator shall consist of all the examinations and tests of this specification performed in the order specified under the paragraph headed Inspection methods and samples submitted for qualification shall be accompanied by the supplier's report showing that the reliability test for the Reliability Qualification Phase (4.4.3.1) has been satisfactorily met on other samples of the same equipment.

4.3.1 Qualification inspection sample - Qualification inspection samples shall consist of three indicators manufactured in accordance with this specification. The indicators submitted for qualification inspection shall have been previously subjected only to the individual inspections. The samples shall be forwarded to the laboratory designated in the Letter of Authorization (see 6.7). A copy of the installation instructions shall be furnished with each Qualification inspection sample in accordance with 3.11.1.

4.3.1.1 Qualification inspection sample identification - The qualification inspection samples shall be plainly identified by durable tags, securely attached, and marked with the following information:

Sample for Qualification Inspection
Indicator, Torquemeter, Engine, Turboprop
Submitted by (Manufacturer's name, date)
for Qualification Inspection in accordance with
Specification MIL-I-22126B(AS) under authorization
(reference letter authorizing tests).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

4.3.2 Periodic Qualification Re-evaluation - In order to retain qualification of products approved for listing on the Qualified Products List (QPL), the manufacturer shall verify by certification to the qualifying activity that his product(s) comply with the requirements of this specification. Unless otherwise specified by the qualifying activity, the time of periodic verification by certification shall be in two-year intervals from the date of original qualification.

4.4 Quality conformance inspection - The Quality conformance inspection shall consist of the Individual inspection, the Sampling plans and the Reliability assurance tests of this specification. The contractor shall furnish all samples and shall be responsible for accomplishing all the inspections. Sampling plan B inspections shall be conducted at a Government laboratory designated by the procuring activity. Quality conformance inspection, except for Sampling plan B, shall be under the supervision of the Government quality control representative. The contractor shall furnish test reports showing quantitative results for all tests required by this specification, signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product.

4.4.1 Individual inspection - Each Indicator submitted for acceptance shall be subjected to the Individual inspection. This inspection shall determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum each Indicator accepted shall have passed the following inspections:

Examination of product
Power-off flag
Range of adjustment
Scale error and friction
Variation in signal frequency
Integral lighting (lighted indicators only)
Response Time
Case Leakage

4.4.2 Sampling plans - The Sampling plans shall consist of Sampling plan A and Sampling plan B inspections. The inspection samples selected for sampling tests shall first have passed the Individual inspections. The inspection samples which have been subjected to Sampling plan A inspection shall not be delivered on contract until they have been refurbished and resubmitted and passed all the Individual inspections. Inspection samples which have been subjected to the Sampling plan B inspection shall not be delivered on contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

4.4.2.1 Sampling plan A sample selection - Sampling plan A samples shall be selected at random in accordance with the following schedule:

Quantity Offered for Acceptance	Quantity to be Selected for Inspection
First 15	(See Note)
Next 50	1
Next 75	1
Next 100	1
Each additional 200 or fraction thereof	1

NOTE: When Sampling plan B is invoked, the quantity shall be zero. When Sampling plan B is to be omitted, the quantity shall be one.

When a defective Indicator occurs, no items from those still on hand or later produced shall be accepted until the extent and cause of failure have been determined, appropriately corrected, and the corrective action approved by the procuring activity. In addition, when a failure occurs, shift to one sample out of fifteen (when Sampling plan B is omitted) and proceed as indicated.

4.4.2.1.1 Sampling plan A inspection - Each sample selected for Sampling plan A inspection shall be subjected to the following tests:

Individual Inspection
Magnetic effect
Voltage and frequency variation
Voltage surge
Power consumption
High temperature operation
Low temperature operation
Indicator response to signal modulation
Vibration resonance

4.4.2.2 Sampling plan B instructions - Three indicators shall be selected at random from the first 15 produced on contract and submitted within 10 days after manufacture. These samples shall be forwarded to a Government Laboratory designated by the procuring activity. Each sample shall be plainly identified by a securely attached, durable tag marked with the following information:

Indicator, Torquemeter, Engine, Turboprop
Submitted by (Manufacturer's name, date)
for production acceptance Sampling Plan
B inspection, of MIL-I-22126B(AS) in
accordance with Contract No._____.
Manufacturer's part number_____.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

4.4.2.2.1 Sampling Plan B approval - Approval of Sampling Plan B instruments shall be by the procuring activity upon satisfactory completion of the designated tests. Any design, material or performance defect made evident during this test shall be corrected by the Contractor to the satisfaction of the procuring activity. Failure of the sample units to pass any of the tests shall be cause for deliveries of instruments under the contract to cease until proper corrective action is approved and accomplished.

4.4.2.2.2 Sampling Plan B inspections - Each sample selected for Sampling Plan B inspection shall be subjected to the following tests:

Salt Fog Test
Sampling Plan A
Vibration cycling
Radio noise interference
Fogging
High temperature exposure
Low temperature exposure
Temperature shock
Endurance
Lamp life (lighted indicator only)
Salt spray
Shock
Test for helium
Internal examination
Lighting contrast (lighted indicator only)

4.4.3 Reliability assurance tests - Reliability assurance tests shall be conducted using MIL-STD-781. Qualification Phase and Production Acceptance (Sampling) Phase Tests shall be conducted. Equipments selected for Reliability assurance tests shall first have passed the Individual Tests.

4.4.3.1 Reliability Qualification Phase - Prior to Qualification Product Listing, QPL, a minimum of three test samples shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of test samples used shall be six. For the Qualification Phase, Test Level E shall be used. The Accept-Reject Criteria for Test Plan I shall be used.

4.4.3.2 Reliability Production Acceptance (Sampling) Phase Tests - Samples of the equipment shall be tested as outlined in MIL-STD-781, under the Section entitled "Production Acceptance (Sampling) Phase of Production Reliability Tests." For the Sampling Phase,Test Level E shall be used. The Accept-Reject criteria for Test Plan II shall be used to determine the length of the tests (until an accept or reject decision is reached).

4.4.3.2.1 Procedure for Production Acceptance (Sampling) Phase - The Reliability Production Acceptance (Sampling) Phase shall be conducted at least once for each lot. Samples shall be selected and placed on test in the same manner as though each quarter of the contract quantity were a lot. The number of samples shall be in accordance with Table 5 of MIL-STD-781. The test program shall begin with the first month's production.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

The samples shall be tested until an accept or reject decision is reached. The test results shall be summarized monthly for the procuring activity. The procuring activity reserves the right to stop the acceptance of equipment at any time after one or more reject decisions have been reached, pending a review of the contractor's efforts to improve the equipment, the equipment quality control, etc.

4.4.3.3 Procedures report and test details - The test details such as the length of the Test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, etc. shall be part of the test procedures to be submitted to and approved by the QPL activity (see 6.7) prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests. The following paragraphs shall be considered as minimum requirements and apply to both phases.

4.4.3.3.1 Duty cycle - The indicator shall be energized during the heating portion of the temperature cycle and supplied with a signal to cause it to be cycled as indicated in Table III. (If a lighted unit is under test, the lighting shall be energized when the indicator is operating).

4.4.3.3.2 Performance characteristics to be measured - The scale error and friction test, described under inspection methods, shall be conducted at least once each week. The indicator shall be observed daily, on regular working days, for any obvious failure such as rough pointer movement.

4.4.3.3.3 Failure criteria - Whenever performance characteristics fall below the acceptance requirements (4.4.3.3.2), at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each should be counted a failure, unless one part caused the other parts to fail.

4.4.3.3.4 Preventive maintenance - No preventive maintenance may be accomplished on the samples while they are on test.

4.4.3.3.5 Operational stability - Adjustments specified in 4.6.24 shall not be required after the Reliability test has been initiated in order to meet specification performance requirements. However, should the total operating time per indicator exceed 1500 hours an adjustment is permitted. But the error prior to adjustment shall not exceed $\pm$40 HP without charging a failure. This adjustment is permitted each 1500 hours of operation of each indicator.

4.4.3.4 Disposition of samples upon completion of tests - Any sample tested may be delivered on contract provided it meets all the following requirements:

(a) It is representative of production units currently being accepted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

(b) It is in "good as new" condition or has been refurbished (see 6.4).

(c) It is otherwise satisfactory.

4.5 Inspection conditions -

4.5.1 Standard conditions - Unless otherwise specified, all inspections required by this specification shall be made under the following conditions:

Temperature Pressure	Room ambient 25° ±5°C Normal atmospheric (approximately 29.92 inches Hg.)
Humidity	Room ambient up to 90% relative humidity
Input Power	115 ±1.0V AC, 400 ±5 Hz. Lighting, 5V in accordance with MIL-L-25467.

4.5.2 Test readings - Unless otherwise specified, before a test reading is taken the indicator shall be vibrated using an electric vibrator set at 30 to 120 Hz with a maximum amplitude of 0.002 inch.

4.5.3 Attitude - Unless otherwise specified, the indicator shall be tested in its normal operating position.

4.5.4 Integral lighting - The indicator lighting system of the lighted indicators shall be energized and shall operate satisfactorily during all tests except humidity, fungus resistance, salt spray and as otherwise specified herein. The lighting shall operate satisfactorily after completion of each test unless otherwise specified herein.

4.5.5 Master test instrument - The master test instrument is a device designed to simulate the output of the torquemeter sensor throughout the range of the sensor. It must be capable of generating the 2 phase signal described in Section 3.4.1 with the phase difference of the given signal known to within ± 1/6 degree and with the frequency of the signal known to ±1%.

4.5.6 Signal frequency - When the frequency of the input signal to the indicator is not specifically called out, it shall be 9210 ±300 Hz.

4.6 Inspection methods -

4.6.1 Examination of product - Each indicator shall be examined externally to determine conformance with the applicable drawings and with all the requirements of this specification and other applicable specifications not covered by tests.

4.6.2 Power-off flag - With no power applied to the indicator, the "OFF" flag shall appear in the window on the dial. With the indicator held in the hand and given a sharp rotational acceleration about its long

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

axis in either direction, it shall not be possible to remove, permanently, the "OFF" flag from the window. Also, with power applied to the indicator the "OFF" flag shall not be visible in the window nor shall the aforementioned manual maneuver be capable of permanently seating the "OFF" flag in the window. The "OFF" flag or the black background shall be correctly displayed in the window throughout all the tests in this specification. This paragraph applies only to indicators required to have a power-off flag (see 1.2).

4.6.3 Range of adjustment - The indicator shall be connected to the master test instrument using interconnecting wiring as shown in Figure 2. The switch on the indicator shall be set to the "B" position and the pointer adjusted to indicate 5700 HP using the "CAL B" adjustment. The indicator switch shall then be set to the "A" position and the "CAL A" adjustment set to the stop which produces the minimum indication. The pointer shall be between -970 and -1025 HP. The "CAL A" adjustment shall then be set to the opposite stop and the pointer shall be between -580 and -640 HP. The pointer shall be observed for smooth movement as the adjustment is operated between its stops. The indicator shall then be adjusted to -800 HP, the switch set to the "B" position, and the "CAL B" adjustment set to the stop which produces the minimum indication. The pointer shall be between 5245 and 5400 HP. The "CAL B" adjustment shall then be set to the opposite stop and the pointer shall be between 5890 and 6045 HP. The pointer shall be observed for smooth operation as the adjustment is operated between stops.

With the indicator switch set in the operate (center) position, the test shall be repeated using the switches in the external wiring.

Insufficient or excessive range of adjustment or erratic movement of the pointer shall be cause for rejection.

The indicator shall then be set to indicate -800 HP with the indicator switch in "A" position and 5700 HP with the indicator switch in "B" position. After making this adjustment, the settings of the adjustment screws shall not be disturbed for the remainder of the tests except when and where noted.

4.6.4 Scale error and friction - The indicator shall be connected to the test instrument and the system energized. The indicator shall then be subjected to a scale error and friction test at room temperature at the test points listed in Table IA or IB. The friction error is the difference in readings made before and after tapping, and shall not exceed $\pm$10 HP at any test point. The scale error is the difference between the observed reading after tapping and the value shown in Table I and shall not exceed $\pm$25 HP at any test point.

4.6.5 Variation in signal frequency - The scale error test shall be repeated at signal frequencies of 6700 $\pm$100 Hz, 10,000 $\pm$100 Hz, and 2700 $\pm$100 Hz at every third test point of Table IA or IB. At the 2700 $\pm$100 Hz frequency the signal amplitudes shall be $0.40 \begin{smallmatrix} + .04 \\ - 0 \end{smallmatrix}$ volts peak-to-peak.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

The scale error at 6700 Hz and 10,000 Hz shall not exceed ±25 HP. At 2700 Hz the scale error shall not exceed ±50 HP and there shall be no oscillation or erratic movement of the pointer. (Note: For this test only, the HP values of Table I apply at each test frequency.)

4.6.6 Integral lighting - Each lighted indicator shall be subjected to and shall meet the requirements of the following tests specified in MIL-L-25467.

(1) Operation

(2) Dial visibility

4.6.7 Response time - The signal input shall be set to produce an indication of 2450 HP. The CAL "A" switch (Figure 2) shall be switched to the open-circuit position. The time required for the indicator pointer to stabilize at -800 HP indication shall be not less than 3 seconds nor more than 7 seconds. The pointer may overshoot, but shall return to the -800 HP indication without oscillation. This procedure shall be repeated using the CAL "B" switch in lieu of the CAL "A" switch. The same requirements shall apply for pointer movement from 2450 HP to stabilization at 5700 HP.

4.6.8 Case leakage - The indicator shall be tested for case leakage by a mass spectrometer type helium leak detector. The mixture of the gases in the case shall have a detected leakage rate from the case, relative to a near vacuum, not exceeding 0.1 micron cubic foot per hour. As an alternate test, an immersion leak test may be performed in accordance with the requirements of MIL-E-5272, Procedure I.

4.6.9 Magnetic effect - The indicator shall be properly connected and power applied. The indicator shall be revolved about a short bar magnet compass with the nearest part of the indicator 5-1/2 inches from the bar magnet. The compass shall have its compensating magnets removed and shall be set up in a uniform magnetic field whose horizontal intensity is between 0.17 and 0.19 oersted. The indicator shall be revolved in a horizontal plane which is perpendicular to the axis of the bar magnet. The indicator shall be held in positions 0, 45, 90, 135, 180, 225, 270 and 315 degrees. At each of these positions the indicator shall be rotated 360 degrees about its horizontal axis. The deflection of the compass at any of the specified positions shall not exceed 5 degrees. This test shall be repeated with no power applied. The same tolerance apply.

4.6.10 Voltage and frequency variation - The indicator shall be energized with a power source that provides the following voltage and frequency combinations: (a) 115V, 400 Hz, (b) 107V, 380 Hz, (c) 120V, 420 Hz, (d) 107V, 420 Hz, (e) 120V, 380 Hz. A scale error test shall be conducted under the conditions of (a), (b), (c), (d), and (e) using every third point listed in Table IA or IB. The scale error under conditions (a), (b), (c), (d), and (e) shall not exceed ±25 HP.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

TABLE I

TESTS POINTS FOR SCALE ERROR AND FRICTION TESTS

For 0° = -800 HP, Range = 6500 HP

A		B	
Set Phase Difference (degrees)	Read HP (Nominal)	Set HP	Read Phase Difference (degrees)
25.	-349	-400	22.2
45.	12.5	0	44.3
65.	374	500	72.0
85.	735	1000	99.7
105.	1096	1500	127.4
125.	1457	2000	155.1
145.	1818	2500	182.8
165.	2179	3000	210.5
185.	2540	3500	238.2
205.	2901	4000	265.8
225.	3262.5	4500	293.5
245.	3624	5000	321.2
265.	3985	5400	343.4
285.	4346	5500	348.9
305.	4707	5400	343.4
325.	5068	5000	321.2
335.	5249	4500	293.5
325.	5068	4000	265.8
305.	4707	3500	238.2
285.	4346	3000	210.5
265.	3985	2500	182.8
245.	3624	2000	155.1
225.	3262.5	1500	127.4
205.	2901	1000	99.7
185.	2540	500	72.0
165.	2179	0	44.3
145.	1818	-400	22.2
125.	1457		
105.	1096		
85.	735		
65.	374		
45.	12.5		
25.	-349		

Note: Tolerances in this specification are given in HP. The following is given for convenience in conversion.

HP	Phase Difference (degrees)
10	0.554
20	1.108
30	1.662
40	2.216

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

4.6.11 Voltage surge - The indicator shall be connected as specified for the voltage and frequency test. The indication shall be set at the beginning of the test at approximately 4000 HP. The voltage and frequency shall be adjusted to 115 ± 1V, 400 ±2 Hz and a voltage surge applied to the power input to provide a momentary voltage of 160 ± 5V (rms) and return to 115 ± 1V. The duration of the surge, including rise and decay time, shall be 0.090 ± 0.005 second. The frequency shall then be increased to 440 ± 2 Hz and a voltage surge applied to provide a momentary voltage of 195 ± 5V (rms) and return to 115V ± 1V. The duration of the surge, including rise and decay time shall be 0.020 ± 0.005 second. Each of the above surge conditions shall be caused to occur five times with at least 15 minutes between surge applications. At the end of this test, the voltage and frequency shall be adjusted to 115 ±1V, 400 ±2 Hz. The indicator shall then be subjected to and shall meet the requirements of 4.6.4.

4.6.12 Power consumption - The indicator shall be properly connected and power shall be applied for a period of five minutes. The power consumption shall be measured at the end of this time and shall not exceed 15 volt amperes for the unlighted indicators, and 20 volt amperes for the lighted indicators. The power factor of the a.c. power circuits shall be within 0.85 lagging to 0.95 leading.

4.6.13 High temperature operation - (This test may be combined with the high temperature exposure test at the discretion of the testing agency.) The indicator shall be properly connected and subjected to an ambient temperature of 71°C (160°F) for a period of 4 hours with power applied. At the end of the 4-hour period and with the temperature maintained at 71°C (160°F) the indicator shall be subjected to a scale error and friction test and the errors for these tests shall not exceed ± 30 HP and ± 10 HP, respectively, at any test point. Not sooner than four hours after return to room temperature the indicator shall be subjected to and shall meet the requirements of 4.6.4.

4.6.14 Low temperature operation - (This test may be combined with the low temperature exposure test at the discretion of the testing agency). The indicator shall be properly connected except that no power shall be applied. It shall then be subjected to an ambient temperature of -54°C (-65°F) for a period of 4 hours. During the last 5 minutes of the 4-hour period power shall be applied to the indicator. At the end of the 4-hour period and with the temperature maintained at -54°C(-65°F) the indicator shall be subjected to a scale error and friction test, and the errors for these tests shall not exceed ± 35 HP and ± 15 HP, respectively, at any test point. With ambient temperature of -54°C still maintained, the power to the indicator shall not be interrupted for a period of 1 hour (total time). At the end of this period there shall be no evidence of any fogging or condensate on the inside of the coverglass. Not sooner than four hours after return to room temperature, the indicator shall be subjected to, and shall meet, the requirements of 4.6.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

4.6.15 Indicator response to signal modulation - This test shall be conducted with an input signal of approximately 185° phase displacement modulated sinusoidally by ± 5.5° at frequencies from 0 to 12 Hz. The response of the indicator to this modulation shall not exceed that shown in Table II. An alternate or additional input signal phase displacement, with the above described modulation, may be selected by the Government quality control representative and the same tolerances shall apply.

TABLE II

INDICATOR RESPONSE TO SIGNAL MODULATION

Modulation Frequency, Hz	Indicator Response, HP
1/2	± 100 MAX. and MIN.
1-1/2	± 100 MAX.
2	± 50 MAX.
3	± 25 MAX.
greater than 4	None Detectable

4.6.16 Aircraft compatibility tests - The qualification inspection sample units shall be installed on P-3 aircraft and tests conducted to determine that the indicator can be calibrated to the engine in accordance with the installation instructions of 3.11.1. Each sample indicator shall be connected to each of the four engines of the aircraft and sufficient calibration checks and engine run-up conducted to demonstrate compatibility of the indicator with the torquemeter assembly. Any incompatibility shall be cause for rejection.

4.6.17 Vibration resonance - The indicator shall be tested in accordance with resonance portion of Procedure XIII of MIL-E-5272. During the entire test the indicator shall be operating at approximately mid-scale. No pointer (either pointer) oscillation in excess of ±10 HP on the scale applicable to the pointer shall be detected during the resonance test. Also, during the resonance test, frequent determination of indicator accuracy at random signal inputs shall be made. There shall be no error larger than ±25 HP. At the conclusion of the vibration test the indicator shall be subjected to and meet the requirements of 4.6.4. There shall be no evidence of wear or damage.

4.6.18 Vibration cycling - The indicator shall be tested in accordance with the cycling portion of Procedure XIII of Specification MIL-E-5272. The indicator shall then be subjected to and meet the requirements of 4.6.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

4.6.19 Radio noise interference - The electrical system of the indicator shall be tested for conducted and radiated radio noise interference in accordance with MIL-STD-462.

4.6.20 Fogging - The instrument shall be energized and placed in a 71° ±2°C controlled ambient temperature for a minimum period of one hour. During this period the indication shall be cycled over approximately 3/4 scale at a rate of approximately one cycle per minute. After this period, and while still at this temperature, an ice cube shall be rubbed on the cover glass for a period of from one to two minutes. The glass shall be wiped dry (do not use compressed air) and the instrument inspected for evidence of water or oil fog inside the cover glass. Evidence of fogging shall be cause for rejection.

4.6.21 High temperature exposure - The indicator shall be tested in accordance with high temperature tests, Procedure II, of MIL-E-5272. At the conclusion of the test and while at a temperature of 71°C, the indicator shall be subjected to the scale error and friction tests. The errors shall not exceed ±30 HP and ±10 HP, respectively, at any test point.

4.6.22 Low temperature exposure - The indicator shall be tested in accordance with low temperature tests, Procedure II, of MIL-E-5272. At the conclusion of the test and while at a temperature of -54°C, the indicator shall be subjected to the scale error and friction tests. The errors shall not exceed ±35 HP and ±15 HP, respectively, at any test point.

4.6.23 Temperature shock - The indicator shall be tested in accordance with Procedure I of MIL-E-5272. At the conclusion of the test the indicator shall be subjected to a scale error and friction test. The errors shall not exceed ± 25 HP and ± 10 HP, respectively, at any test point.

4.6.24 Endurance - The indicator shall be energized and supplied with a signal to cause it to operate in accordance with the test cycle shown in Table III. If a lighted type indicator is under test, the lighting shall be energized when the indicator is operating.

TABLE III

Approx. Value of Applied Signal in HP	Duration of Signal, Minutes
4200	5
3500	10
3000	15
2400	140
Power Off	10

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

The applied signal shall be modulated with an amplitude of about 100 HP at a frequency of approximately 2 Hz. This test shall consist of 300 of these test cycles and at the conclusion of the test, the indicator shall be subjected to and meet the requirements of 4.6.4 except that the error at any test point shall not exceed ±40 HP. The indicator shall then be subjected to and meet the requirements of the range of the adjustment test (4.6.3) and shall be adjusted as described in the final paragraph of that test. Again, the indicator shall be subjected to the scale error at room temperature test and the tolerance of 4.6.4 shall apply. The scale error test following the adjustment may be omitted if the errors during the scale error test following the cycling do not exceed 25 HP.

4.6.25 Lamp life - The indicator shall be subjected to the life test specified in MIL-L-25467. (This test shall be run concurrently with the Life Test if practicable). This test applies to the lighted type indicator only.

4.6.26 Salt spray - The indicator shall be tested in accordance with Procedure I of MIL-E-5272. After completion of the test procedure, the indicator shall be subjected to and meet the requirements of the Individual inspection.

4.6.27 Shock - The indicator shall be subjected to and meet all the requirements of the shock test, Procedure V of MIL-E-5272. The indicator shall be operating with an indication between 2000 HP and 5000 HP during the "Equipment operation" portion of the test and shall be subjected to and meet the Individual inspection requirements, specified herein, following that portion.

4.6.28 Test for helium - The indicator case shall be punctured, or the filling tube cut, and the indicator subjected to the test for helium by means of a mass spectrometer type helium detector. Failure to detect helium shall be cause for rejection.

4.6.29 Internal examination - The case of the indicator shall be opened and the internal mechanism examined. The lamp circuit of the lighted type indicator shall also be examined as specified in MIL-L-25467. Any deterioration or damage which could in any manner prevent the indicator from meeting functional operation or maintenance requirements during service life shall be cause for rejection.

4.6.30 Lighting contrast - The indicator shall be subjected to and meet the requirements of the contrast test specified in MIL-L-25467. This test applies to the lighted type indicator only.

5. PREPARATION FOR DELIVERY

5.1 Packaging - The indicator shall be packaged in accordance with MIL-STD-794 Level A. The method of preservation shall be in accordance with MIL-P-116, Method 1A-5 without preservative compound and using reusable metal containers.

5.2 Packing - The indicator shall be packed in accordance with MIL-STD-794, Level A, B or C as specified in contract or order (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

5.3 Marking - The interior and exterior containers shall be marked as specified in MIL-STD-794.

5.3.1 Precautionary marking - The following precautionary markings shall appear on two opposite sides of each interior package and shipping container whenever practicable.

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

6.1 Intended use - The indicators covered by this specification are intended for use in aircraft to indicate horsepower at 13820 RPM of a turboprop engine equipped with a torquemeter transmitter whose output signal is as described in paragraph 3.4.

6.2 Ordering data - Procurement documents should specify the following:

6.2.1 Procurement requirements -

(a) Title, Number and Date of this Specification.

(b) The quantity, MS part number and type number, if any, of the instrument desired.

(c) Levels of packing (see 5.2) desired.

(d) Government laboratory or testing activity (see 4.4, 4.4.2.2).

(e) That a copy of the installation instructions (see 3.11.1) shall be submitted to the Government laboratory or testing activity with the Quality conformance samples.

6.2.2 Contract data requirements - Data specified in paragraphs 3.11.1, 4.3, 4.4, 4.4.3.2.1, and 4.4.3.3 will be listed directly on a DD Form 1423 incorporated into the contract.

6.3 Drawing requirements - The attention of contractors is invited to the fact that Government contracts for equipment, whether procured directly by the Government or through another contractor, require that all drawings submitted must be in accordance with the requirements of MIL-STD-100.

6.4 Definitions -

6.4.1 Amplitude - Whenever the word "amplitude" is specified, it shall mean the extent of motion as measured from one extreme to the opposite extreme.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

6.4.2 Hermetic Seal - A hermetic seal shall be a perfectly closed and air tight seal made between metallic and metallic or between metallic and vitric materials. A hermetic seal is not intended to include seals accomplished by gaskets.

6.4.3 Good as new - "Good as new" shall mean indicators operated less than 10 percent of the specified MTBF Operation.

6.4.4 Refurbished - Refurbished shall mean that the indicator has been completely overhauled with all component parts meeting current parts standards; and the indicator shall have been subjected to and met all the requirements of a new indicator.

6.4.5 Corrosion Resistance - The terms "corrosion resistant" and "protected as specified herein to resist corrosion" as used in this specification, shall mean a material which exhibits no visual evidence of corrosion subsequent to exposure to the Salt Spray Test, Procedure I of MIL-E-5272. The material shall have been cleaned by ultrasonic methods or equivalent prior to testing to determine corrosion resistance.

6.5 Information supplied with transmitter - The following information is supplied by the transmitter manufacturer with each transmitter, viz:

(a) Torque required to twist the shaft through 360 degrees of phase displacement. This value applies to a specific shaft.

(b) The torque to which a shaft is subjected when it is installed and operating with a running engine under certain specific conditions. This value depends on conditions and is not a function of the shaft. It generally falls in the range of 80 to 220 horsepower at 13820 RPM.

6.6 Spurious signals - The engine accessories, torque shaft gear teeth surface imperfections, etc., are known to produce spurious signals (noise) in the indicator input leads in the aircraft. The voltage amplitudes and frequencies of these spurious signals have not been measured and are therefore undefined. Experience has proven that spurious signals in the frequency range 2.5 KHz to 10.0 KHz are not a major design factor. The manufacturer is cautioned that the indicator is required to be impervious to these spurious signals, therefore, positive noise rejection must be a design consideration.

6.7 Qualification - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for the opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products offered by this

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Department of the Navy, Washington, D. C. 20360, and information pertaining to qualification of products may be obtained from that activity.

6.8 Precedence of documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

DOD Proj. No.
6620-N041

*U.S. GOVERNMENT PRINTING OFFICE: 1972-714-537/311

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION	
MIL-I-22126B(AS)	INDICATOR, TORQUEMETER, ENGINE, TURBOPROP

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*	DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:12Z
Check the source to verify that this is the current version before use.